

PLACEMENT SKILL-GAP HUB

Two-Day Execution Workflow

Hackathon
HACKORBIT 2K26

Team
Codeavengers

Track
Track A — DataDrishti

Institution: Bannari Amman Institute of Technology

How to use this document. This is the execution workflow only — the step-by-step build order, checkpoints, and cut-rules for the 2-day build. For dataset structure, DAX formulas, calculation logic, dashboard content, and judge-defense material, see the companion document **report_HACKORBIT.pdf**.

Guiding rule for both days: build the smallest working version of everything before polishing anything. A complete, plain 3-page dashboard beats a broken, ambitious 4-page one. If a block runs over time, cut scope inside that block — never skip straight to the next block with something half-built.

1. Scope Lock — What We Are Actually Building

This workflow builds the MVP agreed in strategy: one dataset (3 tables), 3 dashboard pages, 10 core DAX measures, and two signature features — the Skill Gap Priority Matrix and the rule-based Training Recommendation table. The branch heatmap is an optional add-on, only attempted if Day 2 is ahead of schedule.

Included in MVP (must finish)	Optional add-on (only if ahead of schedule)
3-table dataset (Students_Placement, Student_Skills, Industry_Skill_Demand)	Branch × Skill heatmap (Page 3 add-on)
Page 1 — Placement Command Center	Extra slicer combinations beyond Branch/Batch/Skill/Status
Page 2 — Skill Demand vs Readiness + Gap table	Company-level skill demand breakdown
Page 3 — Priority Matrix + Recommendation table (Action Center)	Animated/bookmark-based page transitions
10 core DAX measures, each individually verified	Additional advanced DAX beyond the core set

Cut order if time runs short (in this order): 1) branch heatmap, 2) extra slicers/polish, 3) reduce Page 2 visuals to the essentials. **Never cut:** the Priority Matrix or the Recommendation table — these are the two features judges will remember.

2. Day 1 — Build the Analytical Engine

Goal for end of Day 1: Page 1 (Placement Command Center) is fully built, visually clean, and demo-able on its own — even if nothing else existed, Day 1's output should look like a finished mini-project.

Block	Time	Task	Deliverable / Exit Condition
1	Hr 1–2	Design and generate the 3-table demonstration dataset in Excel/CSV. Bake in logical relationships (e.g. Readiness varies by branch, high-demand skills appear more often in the demand table).	Students_Placement, Student_Skills, Industry_Skill_Demand tables exist as clean CSV/XLSX files. Spot-check 5 rows by hand — do the numbers make logical sense?
2	Hr 3–4	Import into Power BI. Power Query pass: fix data types, standardize Branch and Skill spellings, remove blanks/duplicates, trim whitespace.	All 3 tables load without errors. Column data types are correct (numbers as numbers, not text).
3	Hr 4 (cont.)	Build the data model: Student_ID relationship (Students_Placement → Student_Skills, 1-to-many) and Skill relationship (Student_Skills → Industry_Skill_Demand, many-to-1).	Model view shows clean single-direction arrows, no ambiguous/circular relationship warnings.
4	Hr 5–6	Write and test core DAX: Total Students, Students Placed, Placement Rate, Average Package, Highest Package, Company Count, Student Readiness %, Industry Demand %.	Each measure checked against a manual calculation on one filtered branch — numbers match.
5	Hr 7–8	Build Page 1 — Placement Command Center: KPI cards, branch placement comparison, company hiring chart, package distribution. Apply consistent fonts/colors.	Page 1 works standalone: filters respond, no visual errors, looks presentable on a projector.

Day 1 checkpoint (mandatory): before stopping for the night, Page 1 must open cleanly and every filter must work. If it doesn't, that is tonight's actual priority — not starting Page 2.

3. Day 2 — Turn It Into a Winning Product

Goal for end of Day 2: the full 3-page story works live, the two signature features are polished, and the team can present and defend it without notes.

Block	Time	Task	Deliverable / Exit Condition
1	Hr 1–2	Skill Gap Engine: add Skill Gap % measure (Demand – Readiness) and a Priority classification (CRITICAL / HIGH / MEDIUM / LOW) using the agreed thresholds. Build the Page 2 gap table.	Table lists every skill with Demand %, Readiness %, Gap, and a color-coded Priority tag.
2	Hr 3–4	Priority Matrix (signature visual): scatter chart, Readiness on X-axis, Demand on Y-axis, quadrant reference lines and labels. Build this early in the day while the team is fresh — it is the fiddliest visual in the whole build.	Matrix renders correctly, points fall into believable quadrants, labels are readable.
3	Hr 5	Recommendation table: Skill, Gap, Priority, and a short rule-based recommendation sentence per skill (e.g. IF Priority = CRITICAL THEN show intervention text). Label it clearly as rule-based, not AI.	Table on Page 3 (Action Center) shows a recommendation for every CRITICAL/HIGH skill.
4	Hr 6	Visual polish pass: apply the severity color logic consistently everywhere, tidy fonts/spacing, add the four slicers (Branch, Batch, Skill, Placement Status), remove any chart that doesn't answer a specific question.	All 3 pages look like one consistent product, not three separate exercises.
5	Hr 7	Full testing pass on the actual laptop that will run the live demo: click every slicer, try edge cases (a branch/skill with no data), confirm no visual breaks.	Dashboard survives a 10-minute stress test with no errors or blank visuals.
6	Hr 8	Build the 9-slide PPT (official HACKORBIT template), write the 3-minute demo script, and rehearse judge Q&A as a team — each member should be able to explain at least 3 DAX measures unaided.	Full run-through completed at least twice, under time, without checking notes.

Day 2 checkpoint (mandatory): by Hour 6, the Priority Matrix and Recommendation table must both be working. Everything after Hour 6 is polish, testing, and rehearsal — not new features.

4. Risk Watch-List

These are the five most likely reasons a team like ours loses — mapped to exactly where in the workflow each one gets defused.

Risk	Where it bites	Mitigation block
Scope creep — trying to build 4 pages + heatmap + extras in 2 days	Throughout, especially Day 2 afternoon	Scope Lock (Section 1) enforced at every block boundary
Fabricated data mistaken for real placement stats during Q&A	Judge Q&A	Day 1 Block 1 — dataset explicitly generated and labeled as demonstration data; rehearsed phrasing in Day 2 Block 6
Priority Matrix scatter chart fails or looks broken	Day 2 Block 2	Built early (fresh team, full runway) with a fallback: sorted gap table if matrix isn't ready by Hr 5
"Rule-based" logic mislabeled as AI under pressure	Judge Q&A	Explicit labeling in Day 2 Block 3 + rehearsed phrase in Block 6
Team member can't explain a DAX measure when asked	Judge Q&A	Day 2 Block 6 — each member owns and rehearses at least 3 measures

5. Pre-Submission Checklist

- [] Dashboard opens and every visual renders with no errors, live, with no internet dependency.
- [] Every slicer (Branch, Batch, Skill, Placement Status) correctly filters all visuals on its page.
- [] Priority Matrix shows four clearly labeled quadrants with points plotted correctly.
- [] Every number on screen can be traced to a DAX formula that some team member can explain in one sentence.
- [] No one on the team says "AI," "real data," or "prediction" without an accurate qualifier.
- [] The PPT follows the official 9-slide HACKORBIT template — no more, no fewer.
- [] The full demo runs in under 3 minutes without checking notes.

6. 9-Slide Presentation Map

#	Slide	Pulls from
1	Team Details	Cover page of this workflow / team roster
2	Problem Statement	Official HACKORBIT problem statement (report_HACKORBIT.pdf \$2)
3	Proposed Solution	Solution architecture pipeline (report_HACKORBIT.pdf \$3)
4	Innovation & USP	Skill-gap intelligence differentiator (report_HACKORBIT.pdf \$3, \$9)
5	System Architecture	Data model diagram (report_HACKORBIT.pdf \$5)

#	Slide	Pulls from
6	Technology Stack	Power BI, Power Query, DAX — kept deliberately simple
7	Prototype / Implementation Plan	Dashboard pages + this workflow's Day 1/Day 2 blocks
8	Impact & Benefits	Who uses it and how (report_HACKORBIT.pdf §12)
9	Future Scope & Conclusion	MVP vs Future Vision table (report_HACKORBIT.pdf §13)

Keep the deck to these 9 slides. Do not add extra slides for the sake of looking thorough — judges reward a tight, confident story more than volume.

7. Build Status — Review 2 Update

As of Implementation Round submission: dataset, all 3 Power BI pages, both signature features, and the web dashboard companion are built and tested. Every DAX measure and web calculation has been cross-checked against the raw dataset independently — see [report_HACKORBIT.pdf §16](#) for the specific defects found and fixed during that process.

Phase	Status
Strategy & scoping	Complete
Phase 1 — Dataset	Complete — 240 students, 5 branches, 6 skills, 22 sample companies
Phase 2 — Power Query	Complete
Phase 3 — Data model	Complete — 2 relationships, verified single-direction
Phase 4 — DAX measures	Complete — 10 measures, all verified
Phase 5 — Dashboard build (3 pages)	Complete — 1 real bug found and fixed during testing
Phase 6 — Skill-gap intelligence	Complete — Priority Matrix + Recommendation Engine, plus a bonus Branch Skill-Gap Heatmap
Phase 7 — Testing	Complete — 4 real defects found and fixed across Power BI and the web dashboard; see report_HACKORBIT.pdf §16 for the full list
Web dashboard companion	Complete — zero external dependencies, filters unified across Branch/Branch/Skill, MVP vs. Future Scope shown on-page
Phase 9 — Presentation	In progress — use the 9-slide map above and report_HACKORBIT.pdf §14 Judge Defense Q&A